MP204 – Electricity & Magnetism

Problem Set 5

Due in the assignment box by 5pm, Friday 24 April 2015

1. The scalar potential Φ and the vector potential $\vec{A}$ are functions which give the electric and magnetic fields via the formulas

$$\vec{E} = -\vec{\nabla}\Phi - \frac{\partial \vec{A}}{\partial t}, \qquad \vec{B} = \vec{\nabla} \times \vec{A}.$$

In lecture, we showed that these automatically satisfy Maxwell's second and third equations; show that if we impose the Coulomb condition $\vec{\nabla}\cdot\vec{A} = 0$, then Maxwell's first equation can be written as

$$\nabla^2\Phi \quad = \quad -\frac{\rho}{\epsilon_0}$$

and Maxwell's fourth equation can be written as

$$\nabla^2\vec{A} - \mu_0\epsilon_0\frac{\partial^2\vec{A}}{\partial t^2} \quad = \quad \mu_0\epsilon_0\vec{\nabla}\left(\frac{\partial\Phi}{\partial t}\right) - \mu_0\vec{J}.$$

Remember that for any vector field $\vec{V}$, the identity

$$\vec{\nabla} \times \left(\vec{\nabla} \times \vec{V}\right) \quad = \quad \vec{\nabla}\left(\vec{\nabla} \cdot \vec{V}\right) - \nabla^2\vec{V}$$

holds.

Remark: The above versions of Maxwell's first and fourth equations give a way of finding the potentials for a system. If the charge density ρ is known, then the equation for Φ (Poisson's equation) is first solved. Once Φ is obtained, it's put into the second equation, and so knowing the current density $\vec{J}$ gives us a differential equation for $\vec{A}$. This is then solved, and since both Φ and $\vec{A}$ are now known, the electric and magnetic field can be obtained. Although we won't use this technique in this course, those of you who go on to take MP465 will probably use it yourself at some point.

2. An electromagnetic system is described by the scalar and vector potentials

$$\Phi = \frac{2B_0}{\mu_0\epsilon_0}\,t \qquad \vec{A} = B_0\left[x\hat{\imath} + (y\sin\omega t - 3z)\,\hat{k}\right].$$

where B_0 and ω are positive constants.

   (a) Find the electric and magnetic fields in this system.

(b) Show that the potentials satisfy the Lorenz condition

$$\mu_0 \epsilon_0 \frac{\partial \Phi}{\partial t} + \vec{\nabla} \cdot \vec{A} \quad = \quad 0.$$

3. Now consider the following potentials:

$$\Phi = -\frac{3B_0}{\mu_0 \epsilon_0}\, t \qquad \vec{A} = B_0 \left[2x\hat{\imath} - 2y\hat{\jmath} + y \sin \omega t\, \hat{k} \right],$$

   (a) Show that the electric and magnetic fields are the same as in the previous problem.

   (b) Show that the Lorenz condition is no longer satisfied, but the vector potential satisfies the Coulomb condition $\vec{\nabla} \cdot \vec{A} = 0$.

   Remark: The last two problems illustrate the flexibility we have in choosing our potentials. For any given electromagnetic field, there are an infinite number of potentials which give that field, so we can always choose the ones that satisfy some particularly convenient condition, like the Lorenz or Coulomb condition, which will make our calculations easier.

VECTOR CALCULUS FORMULAE IN CARTESIAN COORDINATES

Cartesian coordinates (x, y, z) with constant unit direction vectors $\hat{\imath}$, $\hat{\jmath}$, $\hat{k}$

- gradient of a function $f(x, y, z)$:

$$\vec{\nabla} f = \frac{\partial f}{\partial x}\hat{\imath} + \frac{\partial f}{\partial y}\hat{\jmath} + \frac{\partial f}{\partial z}\hat{k}$$

- divergence of a vector $\vec{A}(x, y, z) = A_x(x, y, z)\hat{\imath} + A_y(x, y, z)\hat{\jmath} + A_z(x, y, z)\hat{k}$:

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

- curl of a vector $\vec{A}(x, y, z) = A_x(x, y, z)\hat{\imath} + A_y(x, y, z)\hat{\jmath} + A_z(x, y, z)\hat{k}$:

$$\vec{\nabla} \times \vec{A} = \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right)\hat{\imath} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right)\hat{\jmath} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)\hat{k}$$

- Laplacian of a function $f(x, y, z)$:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$